

Smart Traffic Lights using Local AI

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Abstract—Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. Recent systems use road information, cameras, and artificial intelligence (AI) to make traffic-signal decisions. This paper discusses the use of local AI for traffic-light control and identifies the need for improved emergency-vehicle response.

I. INTRODUCTION

Smart traffic-light systems are being studied as a way to reduce waiting time and respond more quickly to changing road conditions. Traditional traffic lights usually follow fixed timers. They are simple and inexpensive, but they cannot easily react when one road suddenly becomes crowded. Recent systems use cameras and artificial intelligence (AI) to estimate how many vehicles are present and then adjust the lights. Research reviews also show growing interest in AI-based traffic signal control and its possible benefits for changing traffic conditions [1].

Cloud-based AI systems send camera information to a central server for processing. This can provide strong computing power, but it depends on an internet connection and can introduce communication delay. Local smart-camera systems move some processing closer to the road. This can reduce the amount of information that must be sent over the internet and may allow faster decisions. A small local computer can examine camera input and send a short instruction to the traffic-light controller. Machine-vision research has also explored using camera-based vehicle information to adapt traffic-light timing [2].

II. SYSTEM COMPARISON

The three systems can be compared using decision speed, internet use, and setup cost. Lower decision time means a faster decision.

TABLE I
COMPARISON OF TRAFFIC-LIGHT SYSTEMS

System	Decision speed (s)	Internet use	Setup cost (\$)
Basic Timed Lights	30	Zero	200
Cloud AI Systems	2	High	2,500
Local Smart Cameras	0.5	Low	1,200

III. SIMPLE SYSTEM ARCHITECTURE

The proposed architecture follows four steps. First, Street Cameras capture traffic conditions. Second, the information is sent to Smart Pole Computers, which perform local processing and make a quick traffic decision. Third, selected information can be forwarded to the City Cloud Server for storage and wider analysis. Fourth, the decision is sent to the Traffic Lights, which change their signals.

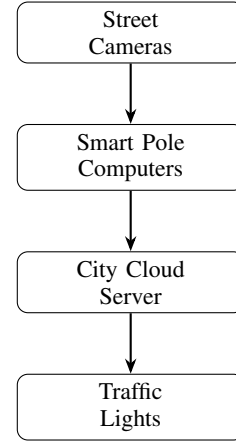


Fig. 1. Architecture of the proposed local AI traffic-light system

IV. RESEARCH GAP

Existing systems work well for regular daily traffic, but they lack a built-in way to automatically clear a path for a speeding ambulance during an emergency. This creates a need for a smart traffic system that can detect an emergency vehicle and quickly coordinate nearby traffic signals to create a clear route.

V. CONCLUSION

The reviewed systems show a common idea: traffic lights can use road information beyond fixed timers to respond to changing traffic conditions. Local processing can support faster decisions while reducing dependence on continuous cloud communication. An important research gap is the automatic coordination of traffic signals to clear a path for an ambulance during an emergency.

REFERENCES

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